

WORKSHEET A.4.M1

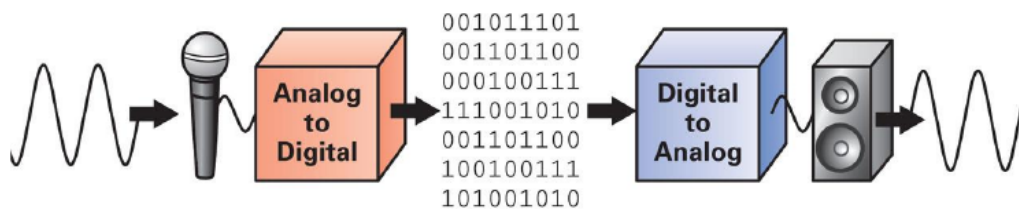
Multimedia - Introduction

What we will learn today:

- ❖ What sound is
- ❖ Lossy and lossless compression algorithms
- ❖ How sound is digitized and the basic characteristics of this process
- ❖ The WAV, MP3 and MIDI standards
- ❖ What an image is
- ❖ Image categories and their characteristics
- ❖ Ways of capturing video
- ❖ Video characteristics and how they affect quality and file size
- ❖ Video compression methods
- ❖ Tools used for video editing

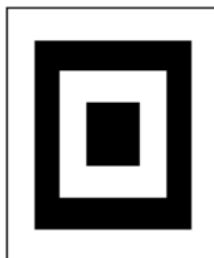
Consolidation Activities

1. What does the image below describe?

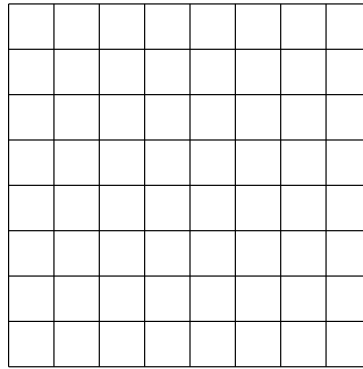


2. Name two image-editing tools.

3. You are given the monochrome image below:



(a) Write the bit values for the image above:



(b) What is the colour depth of the image?

.....

4. Name some ways in which video can be captured.

.....

.....

Assessment Activities

5. What is the difference between the two images below?

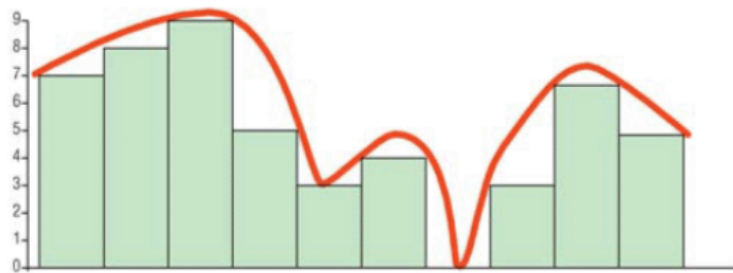


Image A

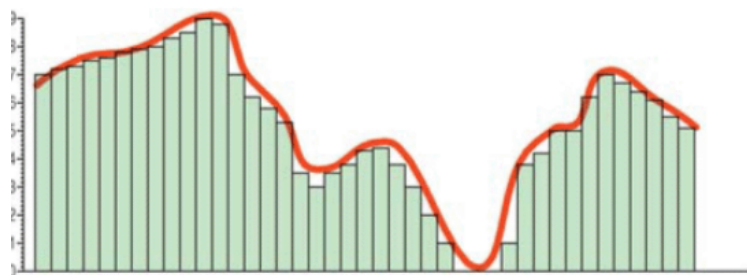


Image B

.....

.....

6. Tick (✓) the options that represent digitized sound formats.

- | | | |
|------------------------------|-------------------------------|-------------------------------|
| ☐ WAV | ☐ GIF | ☐ MP3 |
| ☐ MP4 | ☐ AVI | ☐ JPEG |
| ☐ BMP | ☐ MIDI | ☐ OGG |

7. Open the file A4.1.M1.swf and go to page 2.

(a) What are the values of fps, resolution and colour depth, and which compression method produces the largest file?

.....

.....

(b) How does the fps value affect video quality?

.....

Additional Activities

8. Tick (✓) the options that represent image file formats.

- | | | |
|------------------------------|-------------------------------|-------------------------------|
| ☐ WAV | ☐ GIF | ☐ MP3 |
| ☐ MP4 | ☐ AVI | ☐ JPEG |
| ☐ BMP | ☐ MIDI | ☐ OGG |

9. A video-editing tool allows us to...

.....

.....

.....

Homework Activities

10. Describe the MIDI sound format.

11. At <http://pixlr.com/express/> you can find an online photo editor.

- (a) Select Browse to upload a photograph.
- (b) Apply one of the available effects.
- (c) Save the photograph to your computer.

12. Visit <http://www.powtoon.com/> to create an animation.